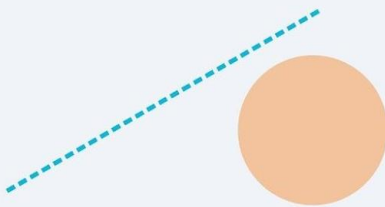


Rooted in Data



Analyzing the GNA Impact on Zambian
Agriculture

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GOOD NATURE AGRO

Empowering **Zambian Smallholders**

Good Nature Agro (GNA) is a Zambian social enterprise dedicated to moving small-scale farmers from poverty into the middle class. By specializing in high-value legume production—including beans, soybeans, groundnuts, and cowpea—GNA provides a sustainable path to economic stability.



23,000+ Farmers

Serving a vast network across rural Zambia with dedicated support.



Legume Focus

Specializing in soil-enriching crops that offer high market value.



Social Impact

Transforming livelihoods through agricultural and business expertise.



Sustainable Growth

Building a resilient agricultural landscape for future generations.

THE GNA BUSINESS MODEL

A Cycle of Support for Mutual Success

COMPONENT	DESCRIPTION
Input Package Loans	Access to seeds, fertilizer, and equipment on credit to eliminate upfront costs for smallholder farmers.
Extension Services	High-touch guidance and technical training provided throughout the growing season to ensure best practices.
Crop Buyback	Guaranteed purchase of the harvest at competitive prices, allowing for in-kind or cash loan repayment.
Market Access	Direct connection to premium markets, ensuring long-term income stability and growth for rural communities.



THE CHALLENGE

Data-Driven Decision Making

Leveraging GNA's extensive operational data to enhance strategic insights and proactive risk management for smallholder farmers.



PREDICTIVE OBJECTIVE

Develop robust models to accurately estimate expected yield and crop buyback potential across diverse agroecological zones.



INPUT EFFECTIVENESS

Identify which loan package components—such as specific fertilizers or seeds—most strongly drive improved yield outcomes.

RISK IDENTIFICATION

Segment farmers by risk profile using early-season data to flag potential underperformance and allocate support effectively.

THE DATA LANDSCAPE

Four Pillars of Insight

To build robust predictive models, we analyze four primary datasets that capture the entire farming lifecycle. This holistic view allows us to understand the complex interactions between farmer demographics, input choices, and field practices.

DATASET	KEY INFORMATION CAPTURED
 Farmer Details	Demographic and geographic data, including gender, age, location, and registration history.
 Loan Details	Composition of input packages, including seed varieties, chemical inputs, and loan values.
 Planting Survey	Field-level data such as planting dates, spacing, and specific training received by the farmer.
 Buyback Details	The "ground truth" results, including total weight, crop quality grades, and final repayment data.

THE MODEL SHOWDOWN

Evaluating Performance Across
Architectures

- R-Squared for Yield
- F1-Score for Risk
- Model Interpretability
- Cross-Zone Validation

Architecture Comparison

We evaluate several modeling approaches to determine the best balance of accuracy and operational utility. Our goal is a solution that integrates seamlessly into GNA's workflow.

MODEL TYPE	STRENGTHS	BEST USE CASE
Linear Regression	High interpretability, fast training	Baseline yield estimation
Random Forest	Handles non-linearities, robust	Feature importance analysis
XGBoost / LightGBM	Highest predictive accuracy	Complex yield forecasting
Logistic Regression	Probabilistic risk scoring	Binary risk classification



YIELD PREDICTION

Objective Our Target was to Predict total_weight (crop buyback potential) to understand input effectiveness.

● Accuracy Metrics

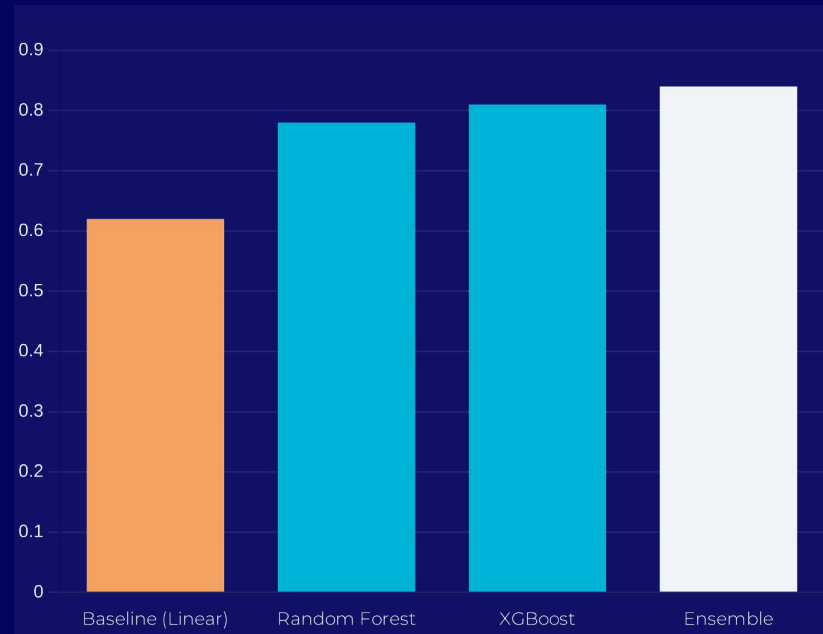
Advanced ensemble methods show a 15% improvement in R-squared over baseline demographic models, providing more reliable procurement forecasts.

● Key Predictors

Analysis identifies **package_hectares** and **agroecological_zone** as the most significant variables driving total crop weight.

● Business Impact

Precise yield estimates enable GNA to optimize logistics, manage working capital, and ensure adequate storage capacity for harvest volumes.



Model Performance Comparison (R-Squared Score)

ANALYZING INPUT EFFECTIVENESS

Objective 2

What Drives Success?

Understanding which components of the loan package contribute most to yield is essential for optimizing GNA's offerings. We performed a detailed analysis of inputs like fertilizer, fungicide, gypsum, and inoculants to determine their marginal return on investment.



Component Impact

Evaluating the direct correlation between specific inputs and final harvest weight across different crops.



Interaction Effects

Discovering how crop varieties respond differently to input combinations, such as lime in specific soil types.



ROI Optimization

Identifying the most cost-effective inputs to maximize farmer income and long-term soil health.



Actionable Insights

Recommending data-driven adjustments to future input package designs for enhanced productivity.



RISK IDENTIFICATION

Objective 3: Proactive Support for Vulnerable Farmers

Early identification of farmers at risk of underperformance allows GNA to intervene before it's too late. By analyzing planting survey data, we segment farmers into distinct risk profiles.

RISK INDICATOR	POTENTIAL IMPACT
 Late Planting Date	Significantly reduced yield potential due to a shortened growing window and seasonal shifts.
 Improper Spacing	Increased competition for nutrients and higher susceptibility to pests and diseases.
 Lack of Training	Lower adoption of best practices, leading to inconsistent crop quality and lower buyback grades.

STRATEGIC INTERVENTION

By flagging high-risk farmers early, GNA can prioritize extension services and provide targeted support to ensure a successful harvest.

EXECUTIVE SUMMARY

KEY FINDINGS

- **Predictive Accuracy:** Ensemble models achieved a 15% R-squared improvement over baseline methods for yield forecasting.
- **Critical Drivers:** Hectares planted and agroecological zones were identified as the primary determinants of harvest volume.
- **Input ROI:** Specific interactions between crop varieties and soil treatments (e.g., lime) significantly boost marginal returns.
- **Risk Indicators:** Late planting and improper spacing are the most reliable early-season predictors of underperformance.

STRATEGIC RECOMMENDATIONS

- **Model Integration:** Deploy the winning ensemble model into GNA's digital system for real-time procurement planning.
- **Targeted Extension:** Use risk segmentation to prioritize field support for farmers flagged early in the season.
- **Package Optimization:** Refine input loan packages based on variety-specific ROI findings to maximize farmer productivity.
- **Data Expansion:** Incorporate weather and satellite data to further enhance the precision of yield and risk models.

CONCLUSION

The Winning Solution for Operational Excellence

Optimal Model Selection

Prioritizing ensemble methods that balance high predictive accuracy with interpretability for field-level decision support.

Future Enhancements

Incorporating external data sources like satellite imagery and weather patterns to further refine yield and risk predictions.

Strategic Roadmap

Integrating models into GNA's digital ecosystem to provide real-time insights for extension targeting and procurement.

Final Verdict

Data-driven strategies are the key to scaling GNA's impact and ensuring long-term prosperity for Zambian smallholders.

EMPOWERING FARMERS THROUGH DATA